

Number Patterns & Tables

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS An **input-output table** follows a **rule**. The rule tells what to do to each input to get its output. Look at how each input changes into its output to **find the rule**. Example: in 1, 2, 3, 4 → out 3, 6, 9, 12, the rule is $\times 3$. Once you know the rule, you can **extend the pattern** (find the next outputs) and **describe the rule in words**.

Round 1 — Retrieve It

1. What does the rule of a table tell you to do?

2. In the table in 1, 2, 3, 4 → out 3, 6, 9, 12, what is the rule?

3. Using that rule, what is the output when the input is 5?

4. After you find the rule, what can you do with it?

Round 2 — Sketch It

Make a two-column table with the heads In and Out. Fill in the inputs 1, 2, 3, 4, 5 and the outputs using the rule $\times 3$.

In	Out
1	
2	
3	
4	
5	

Number Patterns & Tables — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A table shows in 2, 3, 4 → out 6, 9, 12. Find the rule and write it.

2. A table shows in 1, 2, 3, 4 → out 5, 6, 7, 8. Find the rule and find the output when the input is 10.

3. Extend this pattern by writing the next three numbers: 4, 8, 12, 16, ____, ____, ____. What is the rule?

Round 4 — Reason from Evidence

For in 1, 2, 3 → out 3, 6, 9, a student said the rule is "+ 2" because $1 + 2 = 3$. Explain why that rule does not work for the other rows, and give the correct rule.

Number Patterns & Tables — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the steps you use to find the rule of an input-output table. Use the words input, output, and rule.

C**Connect**

Connect a rule to a real pattern, like the number of wheels on 1, 2, or 3 cars. What is the rule and what does it stand for?

E**Extend**

Make your own table with the rule $\times 4$. Fill in 4 rows, then describe the rule in words.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Number Patterns & Tables

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. What to do to each input to get its output.
2. $\times 3$ (multiply the input by 3).
3. 15 (5×3).
4. Extend the pattern (find more outputs) and describe the rule in words.

Sketch It — look for

A two-column table, In: 1, 2, 3, 4, 5 and Out: 3, 6, 9, 12, 15 (each output is the input $\times 3$).

Apply It

1. Rule: $\times 3$ (each output is the input times 3).
2. Rule: $+ 4$ (add 4 to the input); when input is 10, output is 14.
3. Next three: 20, 24, 28; rule: $+ 4$ (or count by 4s / $\times 4$ from the position).

Reason from Evidence — look for

" $+ 2$ " only works for the first row. For input 2 it would give 4, not 6, and for input 3 it would give 5, not 9. The rule that works for every row is $\times 3$.

ACE — Extend look-for

A correct table for the rule $\times 4$ (for example, In 1, 2, 3, 4 $\rightarrow$ Out 4, 8, 12, 16) with the rule described in words as "multiply the input by 4."

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.